

SUMMER INTERNSHIP REPORT (EC68221)

On

Diagnosis of Dengue through FTIR and Raman Spectroscopy

(To build a Machine Learning Model for efficient diagnosis)

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Submitted by:

Batta Chaitanya

(21EC10012)

Under the supervision of

Prof. Basudev Lahiri

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Abstract

This report presents an account of my Machine Learning Engineering (MLE) Internship at Electronics and Electrical Communication Department, Indian Institute of Technology Kharagpur under Prof. Dr. Basudev Lahiri in the summer of 2026. The primary focus of the internship has been to build an efficient ML model that can diagnose Dengue infection through FTIR Spectroscopy Data.

The Data was sourced from the work done by S Das et al, 2025. This report and during the internship, the actual data I had received had been the first derivative transformed data of the original spectral intensity data.

This report tries to show educated guesses from the plotted data that motivate further transformations to the Data. It also shows the guardrails such as penalised class weights that were used to train the data from hazardous misclassifications.

Statistical Variability of the Data across classes had been checked for different transformations and it was found through t-SNE visual test that second derivative transformation data is more efficiently classifiable than the first derivative transformation data.

ML models and classifiers such as Support Vector Machines, Random Forest, XGBoost, LightGBM, Multi Layer Perceptron and Neural Networks have been tested for both data to assess which of these models can produce better results and for which data.

Standard metrics like Accuracy, Precision, F1 score and recall have been used during the training and testing of the model for assessment of a better model for the dataset. Critical misclassification as a metric is also included, that is if an infected sample was predicted to be healthy to keep track of hazards during training and testing of the data.

MLP and Neural Networks performed the best given all the metrics providing around 94% accuracy for classification into the 4 classes and 0 critical misclassifications.

Acknowledgement

Foremost, I am incredibly grateful to Prof. Basudev Lahiri (Faculty Advisor and Project Guide), Prof. Tarun Kanti Bhattacharya (HoD, E&ECE) and Prof. Arun Chakraborty (Dean of Student Wellbeing) for providing me this opportunity to complete my internship at E&ECE department of IIT Kharagpur. It is only because of their express support and encouragement, have I been able to do this internship successfully.

My sincere thanks to my seniors and mentors Subhanita Roy and Souvik Das for guiding me through difficult parts of the internship by thoughtful suggestions, insightful comments and corrections. I appreciate their helpful attitude towards me in my time of need.

I also thank everyone from the Institute fraternity who stepped up for me as their family.

Introduction

Dengue is a vector borne infection caused by the bite of female Aedes mosquitoes which are the hosts for the virus commonly labelled as DENV. This disease is mostly endemic to tropical and subtropical regions, placing the Indian subcontinent right in the zone prone for infections. (S. Das et al, 2025)

WHO reported about 14.6 million infections of DENV with fatal infections near 12000 in the year 2024. Every year 100 million people across the world are at a risk of DENV infection. India alone reported 121,824 cases in the year 2025 with 131 confirmed dengue related deaths. (WHO, 2025)

Treatment for DENV infections is available and early intervention helps deal with the virus better. Early intervention requires early detection. Enzyme Linked Immunosorbent Assay(ELISA) is a serological test that can detect Non-Structural Protein 1(NS1) in the first 7 days of infection.

NS1 is a protein the DENV produces in large quantities in the initial stages of infection. (Mahmood T et al, 2018) The body starts producing antibodies like IgM and IgG at different stages to fight back against the infection. IgM starts appearing a few days after the initial infection and can last a couple of weeks. IgG presence usually means a secondary or prior infection.

ELISA is a powerful tool for the diagnosis of infections such as Dengue but it usually takes 24-48 hrs to get a comprehensive report. There are also Rapid Detection Tests(RDTs) but are often quite expensive and require trained personnel for diagnosis.

FTIR and Raman Spectroscopy for Diagnosis

Fourier Transform Infrared (FTIR) and Raman Spectroscopy are methods used in combination to assess the molecular composition of a certain sample. They rely on absorption and scattering properties of light in reaction to molecular composition of the samples.

These methods can detect the presence of molecular bonds that correspond to NS1 protein and also the antibodies IgM and IgG. Thus, giving us an opportunity at a possible diagnostic for DENV.

Data Origin

To this end, a total of 302 samples, of which 142 were DENV infected, 120 were infected with Chikungunya and 40 healthy samples were collected by S Das et al, 2025.

The DENV samples, which shall be my main focus have been further tested through ELISA for the presence of NS1, IgM and IgG and labelled accordingly before Spectroscopy along with the healthy samples and obtained the Spectral Intensity Data across 680 wavenumbers.

Although my main motive is to build a model that can diagnose a DENV infection, I have tried to make the model identify different types of antigens and antibodies as well.

It was also in this paper that the second derivative of the spectral intensity data could be used for classification with a machine learning model.

Objective:

To build a multiclass classification model that can efficiently identify infected samples from the healthy ones to facilitate diagnosis of Dengue through FTIR Spectroscopy

Data:

The dataset received was the first derivative of FTIR Spectral Intensity Data of 160 samples across 680 wavenumbers. I also further took another derivative of the received data(that is the second derivative overall).

1. 1st deriv. Data(160 samples * 679 features)
2. 2nd deriv. Data(160 * 678)

The data received had 5 different classes, 'Dengue_IgG'(24 samples), 'Dengue_NS1'(71), 'Healthy_ATR'(40), 'Dengue_IgM'(4) and 'Dengue_NS1+IgM'(21).

The samples had been labelled as one of the above classes after prior testing with ELISA.

Preprocessing and Labelling:

Since Dengue_IgM has only 4 samples related to it, it might be underrepresented in the model. So I merged Dengue_IgM and Dengue_NS1+IgM into a single 'Dengue_IgM_Combined'(25 samples).

I had implemented class weights to penalise the misclassification of infected samples as Healthy wherein if such mislabelling is found, the weight of that class would increase by 1.5 times.

Customized class weights (adjusted for cost-sensitive learning):

Dengue_IgG (encoded: 0): 2.53(1.68)

Dengue_IgM_combined (encoded: 1): 3.20(1.60)

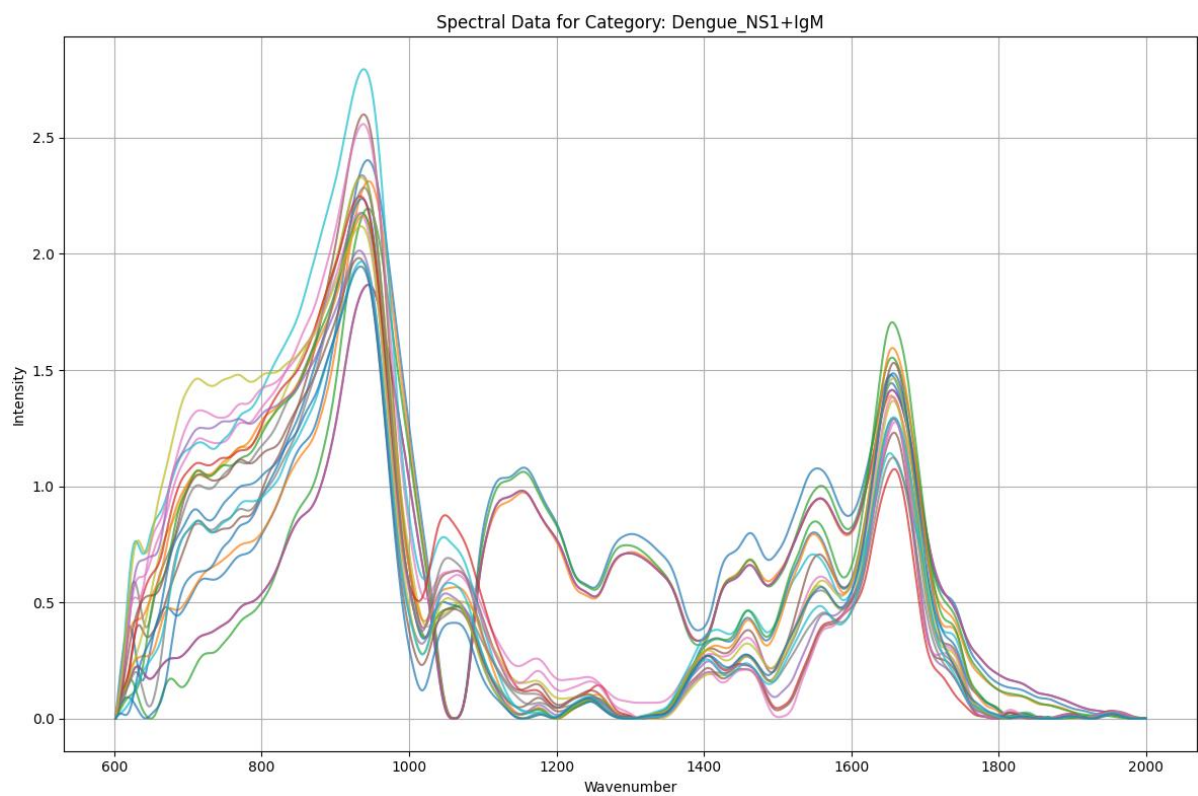
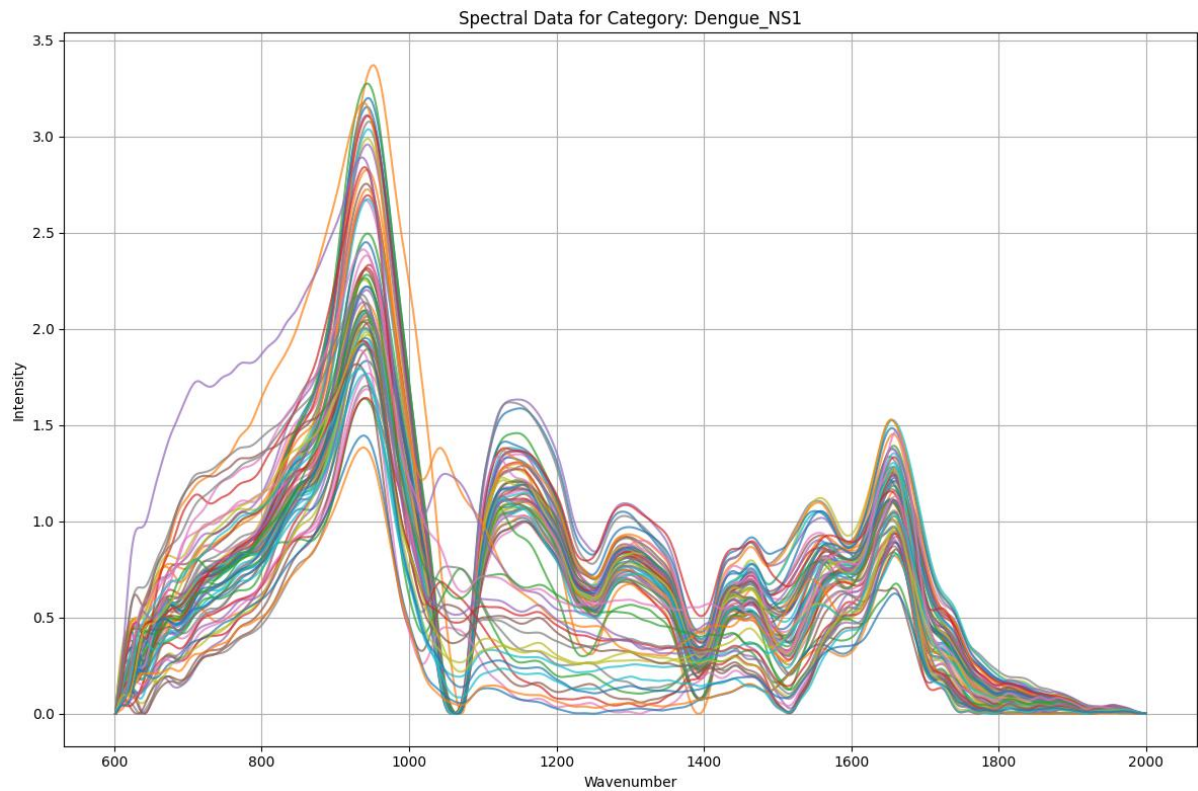
Dengue_NS1 (encoded: 2): 0.84(0.56)

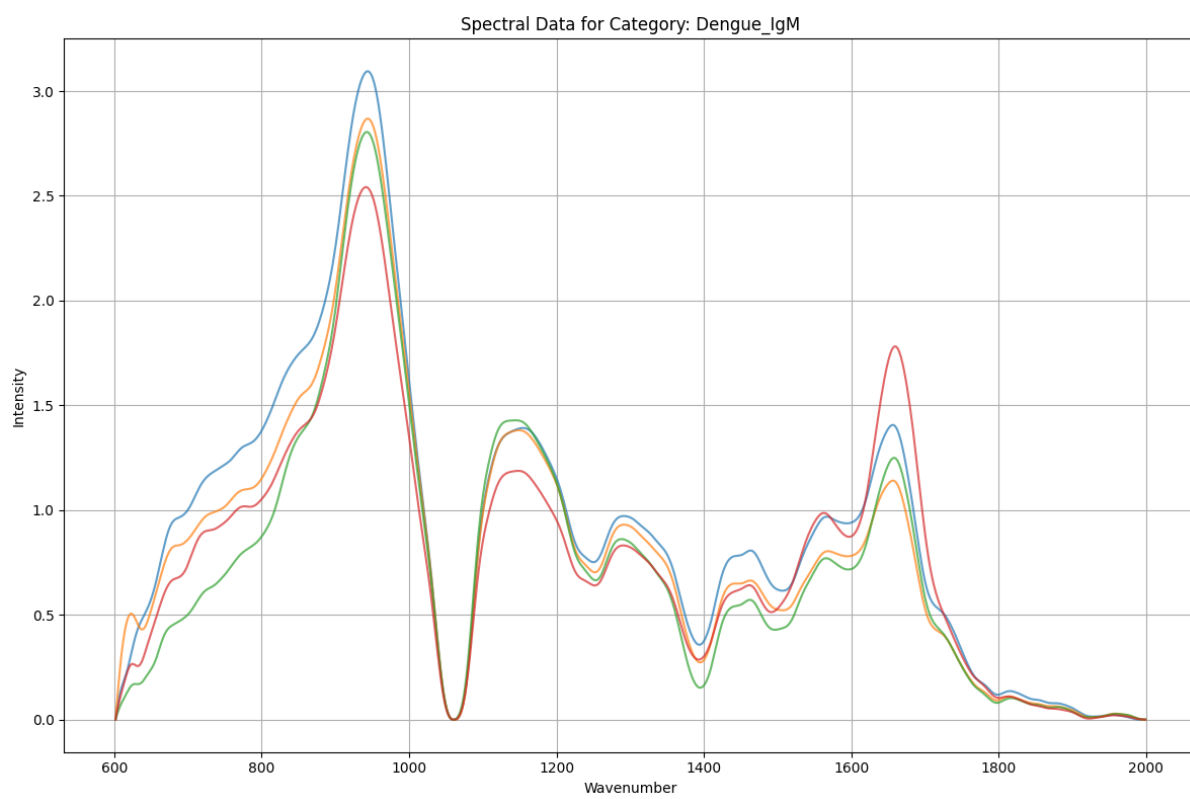
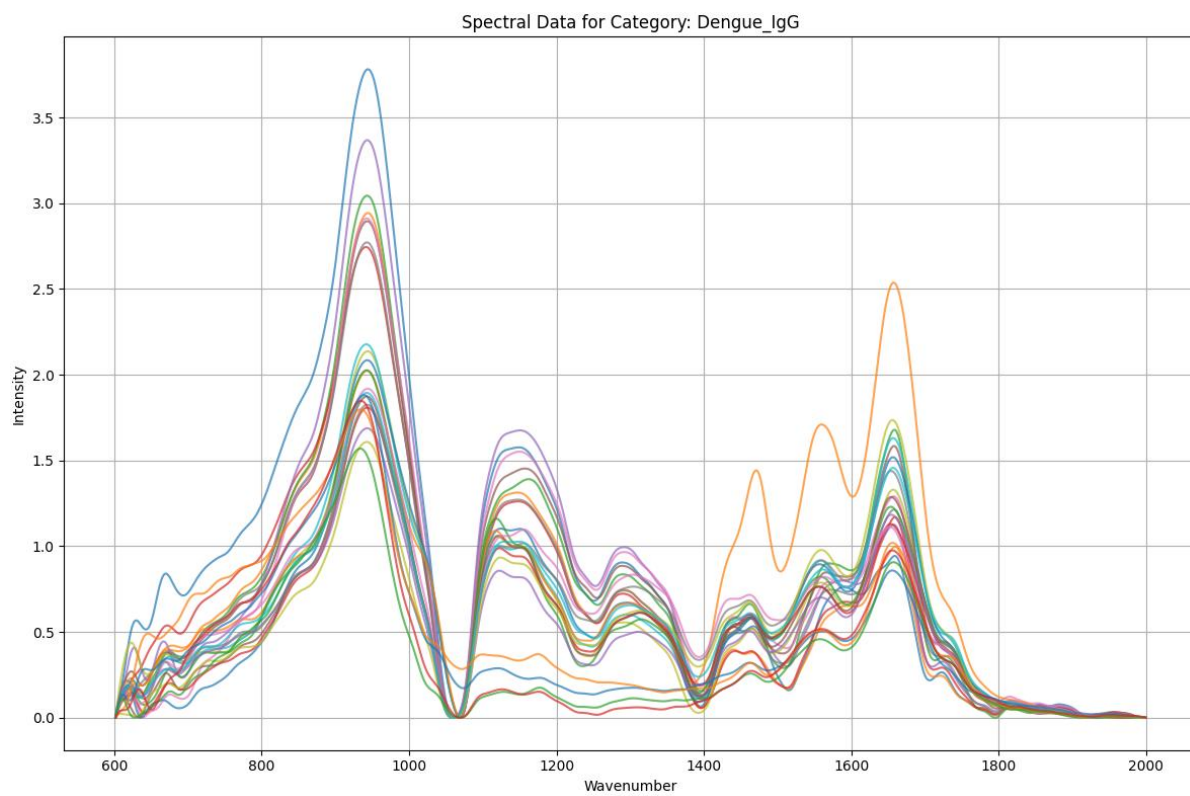
Healthy_ATR (encoded: 3): 1.00(1.00)

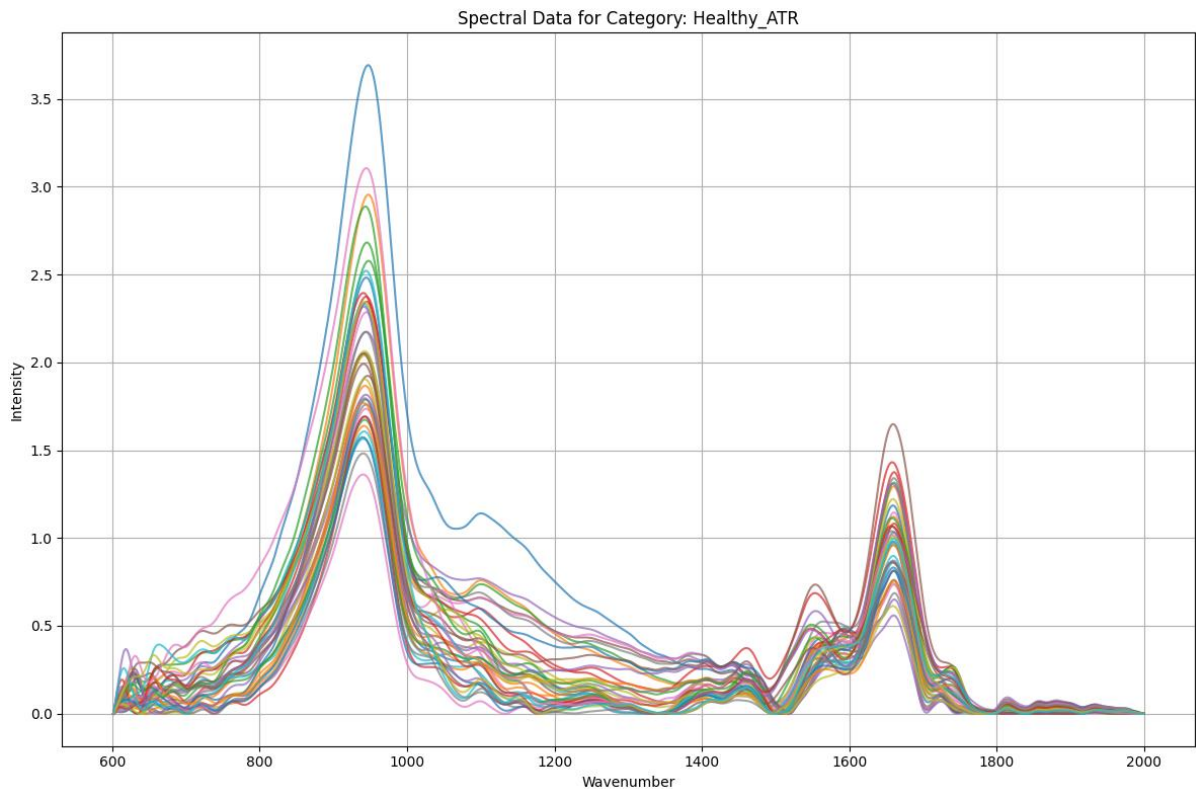
Note: the values inside the parantheses are the initial balanced class weights.

Plots:

1st deriv Data





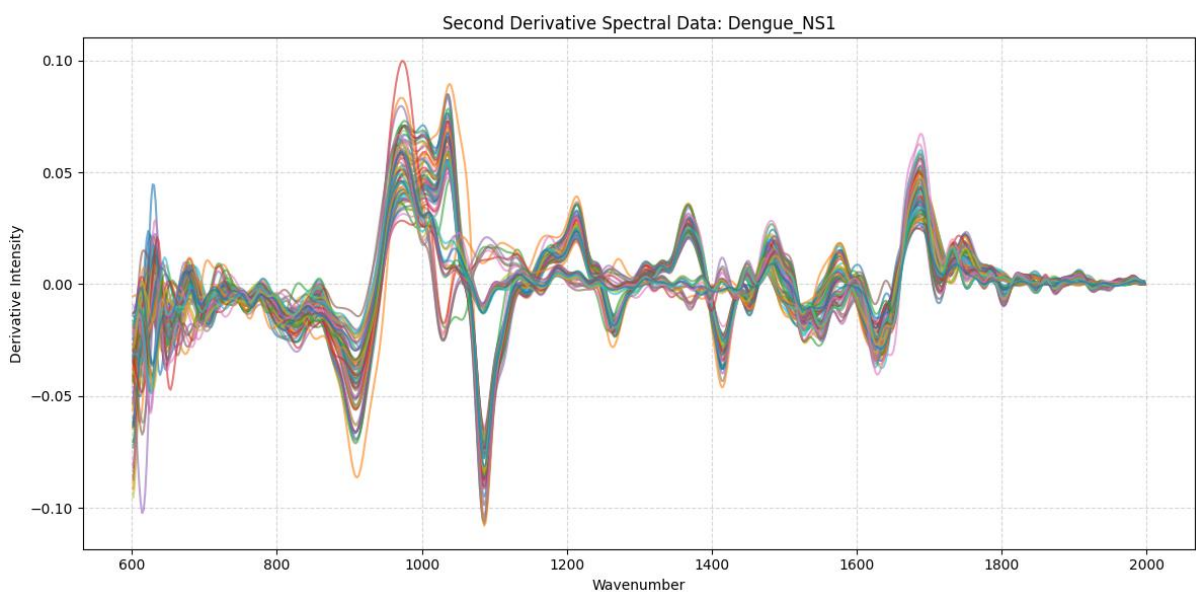


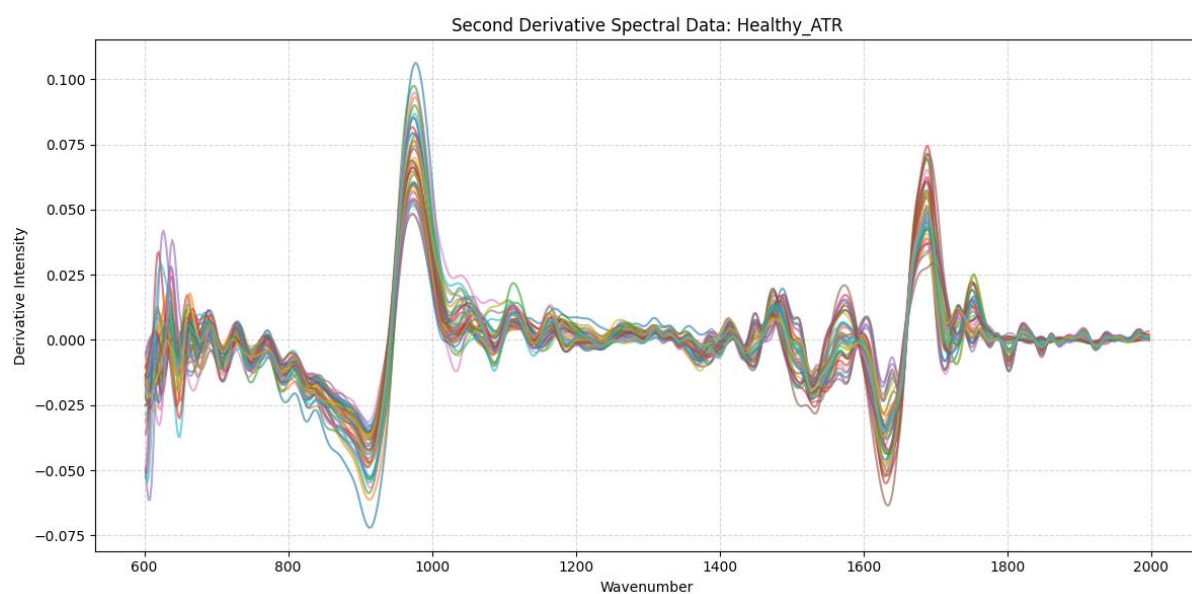
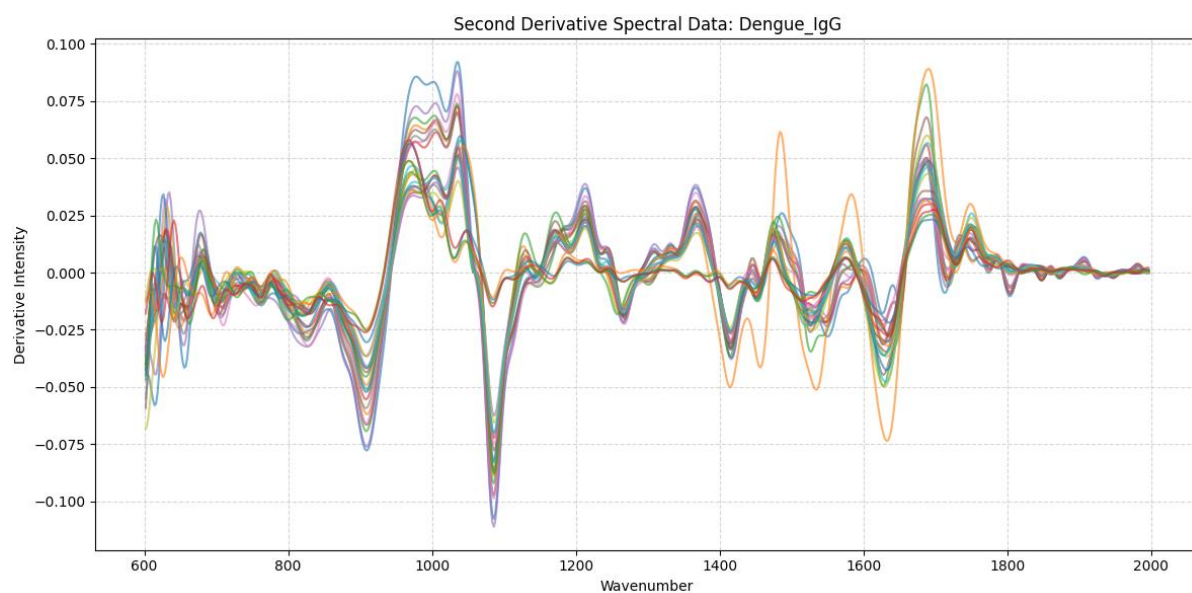
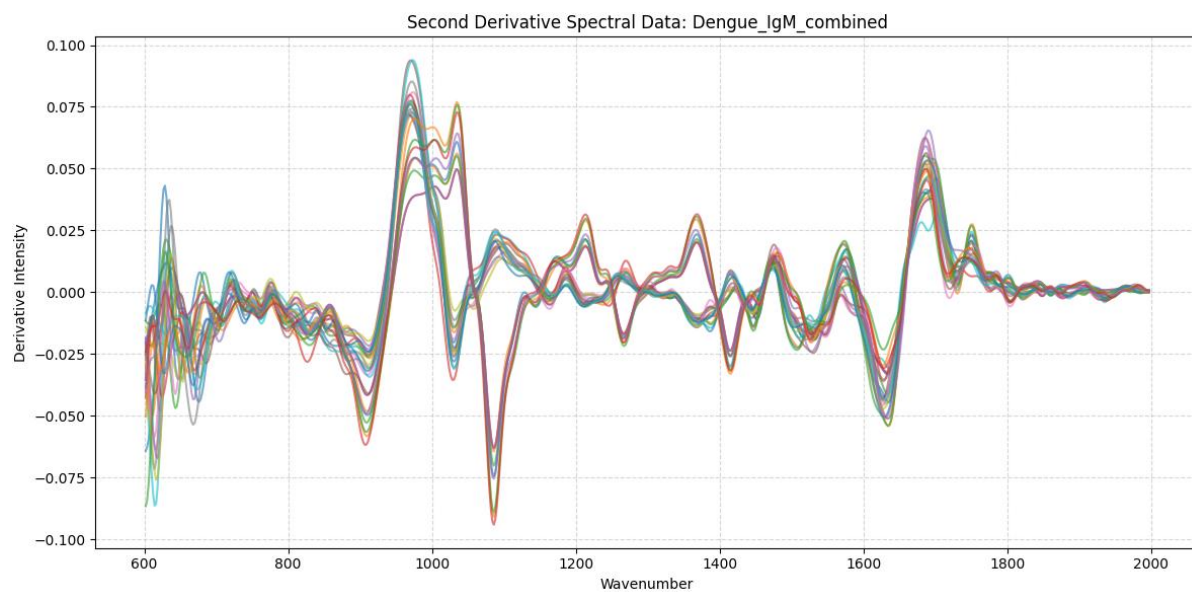
From these plots, we observe variability across each class but especially between the healthy samples and infected samples in the region between 1000cm-1 to 1200 cm-1. Healthy samples appear to lack a minimum whereas most other samples have minima in the region.

This is what motivated me to take another derivative of the received data, as a minimum after taking a derivative would yield a strong zero in the region, which might help in classification.

Here are the

2nd deriv Data plots





There is a strong zero in all the plots except the healthy samples in 1000 to 1200cm⁻¹ region

Statistical Analysis

The assumed features might be noise statistically. So I wished to check the statistic variability among the classes from the sample data. t-Distributed Scholastic Neighbour Embedding (t-SNE) is a great visual tool that reduces the dimensionality of the Data into 2D or 3D representations for better observation of variability in Data.

So, I had produced tSNE scatter plots for both the 1st and 2nd deriv. Data

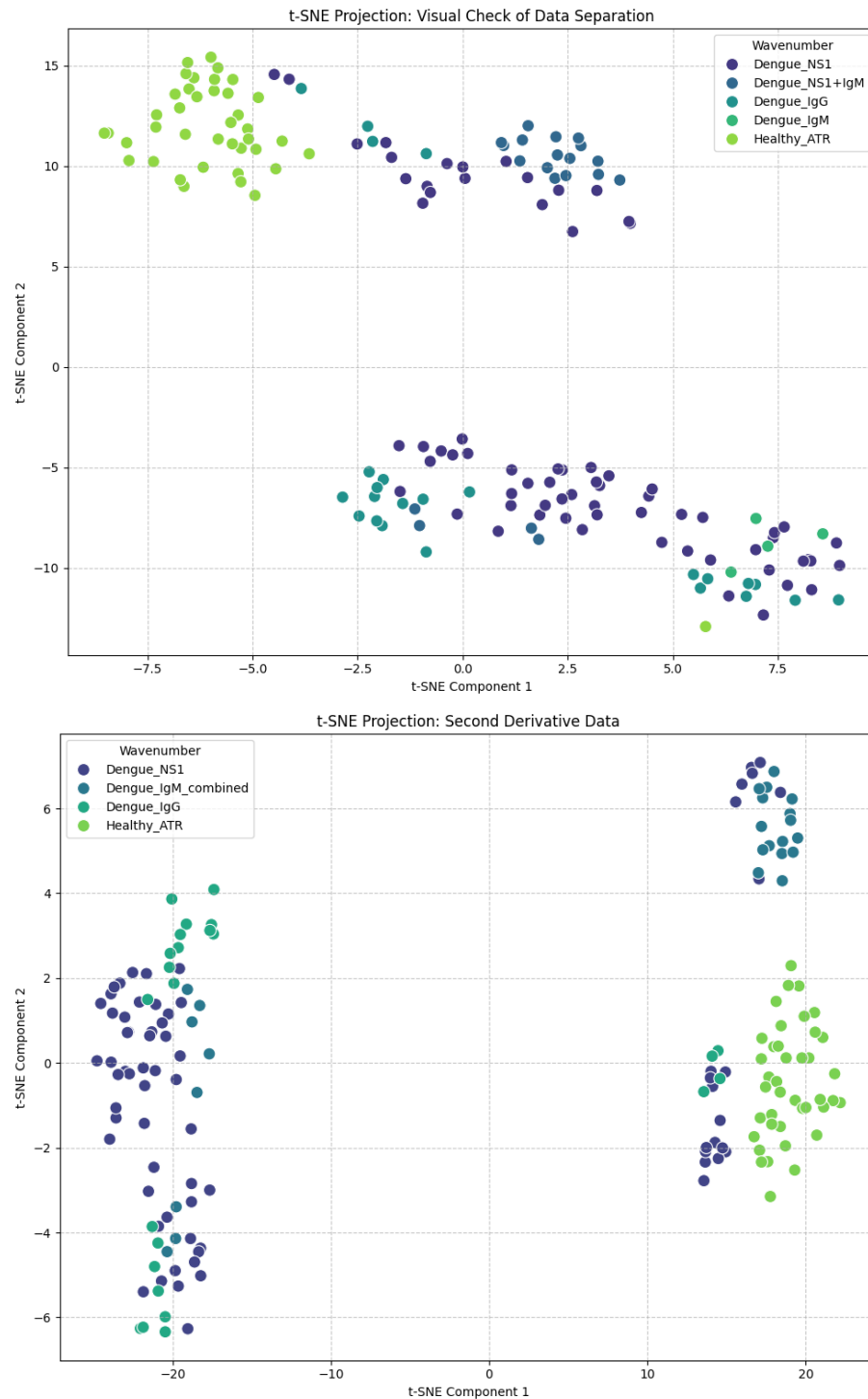


Fig. t-SNE plots for first and second derivative transform data

In both plots, we see a clear clustering of Healthy samples near each other whereas the infected samples cluster but are not as classifiable from each other as they are from the healthy ones.

We can observe a single healthy outlier in the 1st deriv data but none in the 2nd deriv data. Also the clusters seem to be more structured in the 2nd deriv. data.

I believe that 2nd deriv data would produce a much clearer classification than the 1st deriv data.

Models and Performance:

Model Architecture	Feature Engineering	Accuracy	F1-Score (Healthy)	F1 (Disease Avg)	Critical Misclassifications
Random Forest (Baseline)	First Derivative	0.78	0.80	0.74	2
Random Forest	Second Derivative	0.88	1.00	0.79	0
SVM	Second Derivative	0.75	0.93	0.69	1
XGBoost	Second Derivative	0.84	0.93	0.80	1
LightGBM	Second Derivative	0.81	0.80	0.77	2
MLP (Scikit-Learn)	Second Derivative	0.94	1.00	0.91	0
Keras (Baseline)	First Derivative	0.81	1.00	0.74	0
Keras (Deep Learning)	Second Derivative	0.94	1.00	0.93	0

Table 1

Accuracy = $\frac{\text{True Positives (TP)} + \text{True Negative (TN)}}{\text{TP} + \text{TN} + \text{False Positive (FP)} + \text{False Negatives (FN)}}$

Precision = $\frac{\text{TP}}{\text{TP} + \text{FP}}$

Sensitivity (Recall) = $\frac{\text{TP}}{\text{TP} + \text{FN}}$

F1-Score = $\frac{2[\text{Precision}][\text{Recall}]}{[\text{Precision} + \text{Recall}]}$

Results and Discussion:

As expected, First derv data had poorer performance across most metrics as compared to the second derv data as we can observe between Random Forest models on both the data.

Then after Random forest, I had tried to gauge the classification in a few non linear classifiers but had been running into issues relating to infected samples being mislabelled as Healthy, which is big hazard.

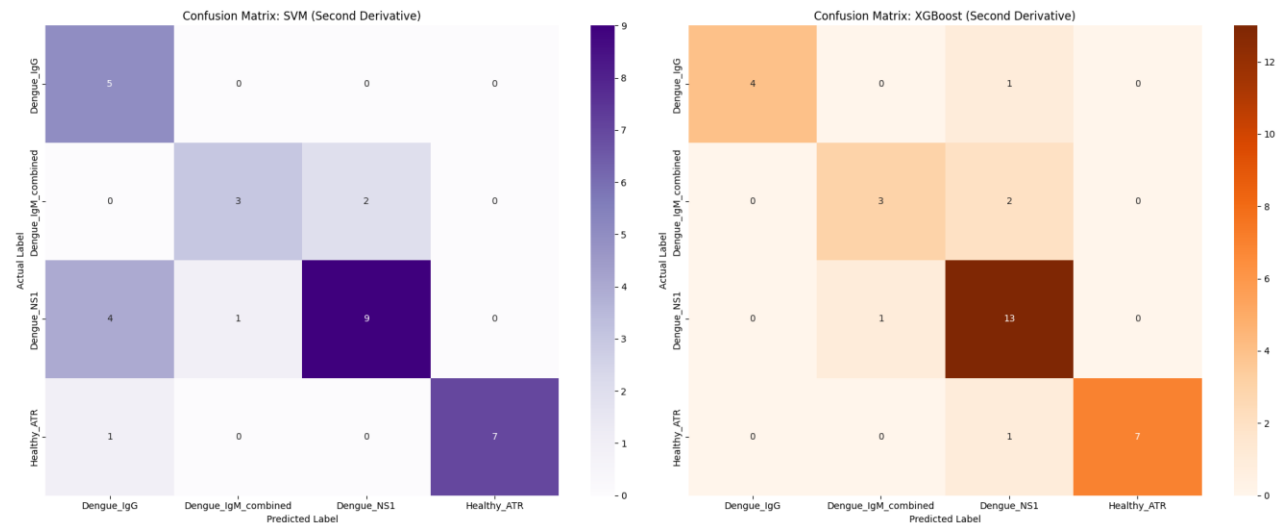


Fig. Confusion Matrices for SVM and XGBoost

That is when I tried to classify using neural networks, starting with Multi Layer Perceptron(MLP) and zero hazardous misclassifications occurred along with an unprecedented boost in accuracy and other metrics. Similarly for Keras Neural Network.

Keras Model: 95,556 parameters.

Input (678) > Dense (128) > Dense (64) > Output (4).

Scikit-Learn MLP: 34,204 parameters.

Input (679) > Dense (50) > Output (4).

These models used Cross Entropy Loss optimizing Accuracy with Adaptive Moment Estimation(adam) optimizer.

Despite the Keras model performing almost as good as MLP model and slightly better, Keras neural network used like 3 times as many parameters.

The best performing model here is Keras neural network on 2nd derv data, but there might be issues in scaling this. So, for bigger, more testable data MLP might be a better option.

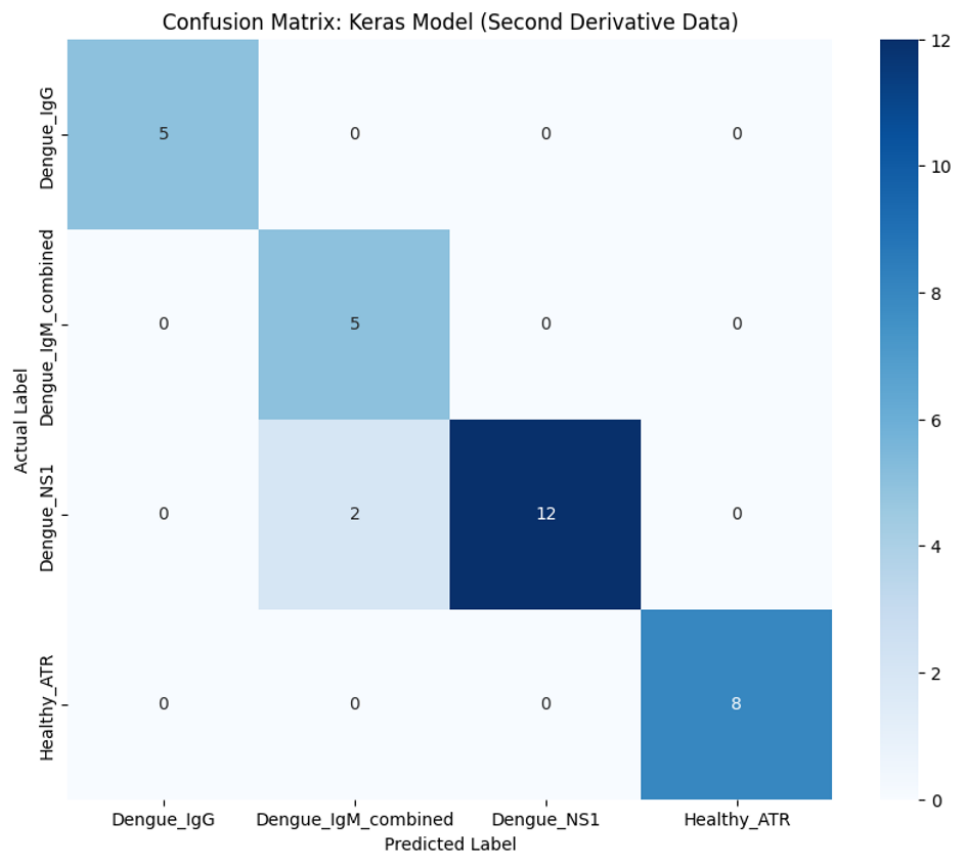


Fig. Confusion Matrix for Neural Network

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